ECE Study Assistant - capstone report

A LangGraph retrieval-augmented agent over eight lecture-note documents. Tasks 4.1-4.4.

Run configuration: **offline** generation, **sentence-transformers/all-MiniLM-L6-v2** embeddings, chunk 800/150, top-k 4. Generated 2026-10-11. Reproducible with no API key.

1. Overview

The system answers three kinds of question: those grounded in the notes (retrieve, grade, answer with a citation), those needing a calculation or formula (a tool call), and small talk. It runs with no API key using a deterministic extractive generator and upgrades to Groq or Gemini when a key is present. The graph is deliberately conditional and cyclic - a failed relevance grade rewrites the query and retries retrieval - which is the mechanism the evaluation measures.

2. Retrieval (4.1)

Pipeline: PyPDFLoader -> RecursiveCharacterTextSplitter(800/150) -> all-MiniLM-L6-v2 (384-d, L2-normalised) -> FAISS IndexFlatIP. The corpus is 8 generated documents, 27 chunks, mean 652 chars. Flat inner product over normalised vectors gives exact cosine similarity, so the score in every trace is a real cosine in [-1, 1].

3. Agent workflow (4.2)

The agent is a LangGraph state machine with a MemorySaver checkpointer. Routing is a runtime decision and a failed grade loops back through query rewriting; a linear chain cannot express that. The LLM layer exposes four decisions (route, grade, rewrite, generate) rather than a generic prompt call, which is what lets the whole system run and be evaluated with no key. Tools are offline: a calculator behind a restricted AST walk (never eval) and a curated, sourced formula table.

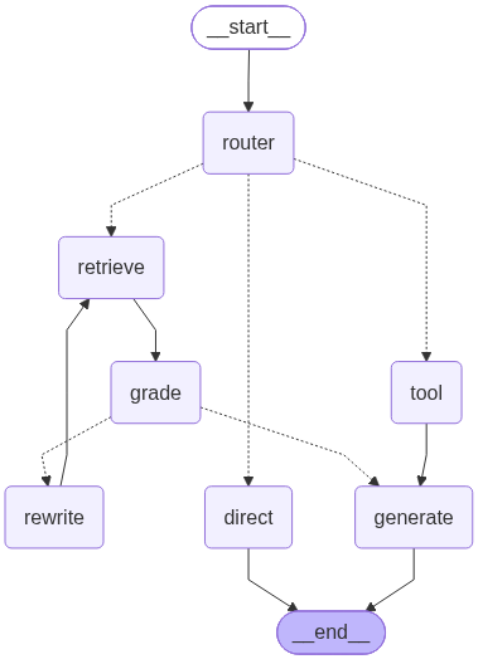


Figure 1. Topology, rendered from the compiled graph.

4. Evaluation (4.3)

22-question labelled set (offline mode). Latency median 26.0 ms, p95 31.1 ms. LangSmith traced remotely: False.

Metric	Value
routing accuracy	100%
retrieval hit rate	100%
answer keyword rate	93%
tool success rate	100%
out-of-scope refusal rate	67%

The three 100% metrics are a ceiling from a small, self-authored set - the two below 100% are the informative results. **q20 (Friis equation)**: absent from the corpus, but 'transmission ... formula' passes lexical relevance grading, so the agent gives a fluent, cited, wrong answer - the case for a generative grader. **q22 (FM bandwidth)**: retrieval was correct; the extractive generator quoted the neighbouring sentence instead of Carson's rule - a generation failure, not a retrieval one.

4.1 Ablations

chunk / overlap	chunks	hit@4	top-1 cos
400 / 100	48	100%	0.593
800 / 150	27	100%	0.553
1200 / 240	17	100%	0.517

Retry policy: the low-coverage floor drops unanswerable questions from **8.0 to 4.0 nodes** and **49.0 to 22.5 ms**, with refusal correctness unchanged. Chunk size is a flat curve (hit@4 saturated); smaller chunks sharpen top-1 similarity at the cost of a larger index.

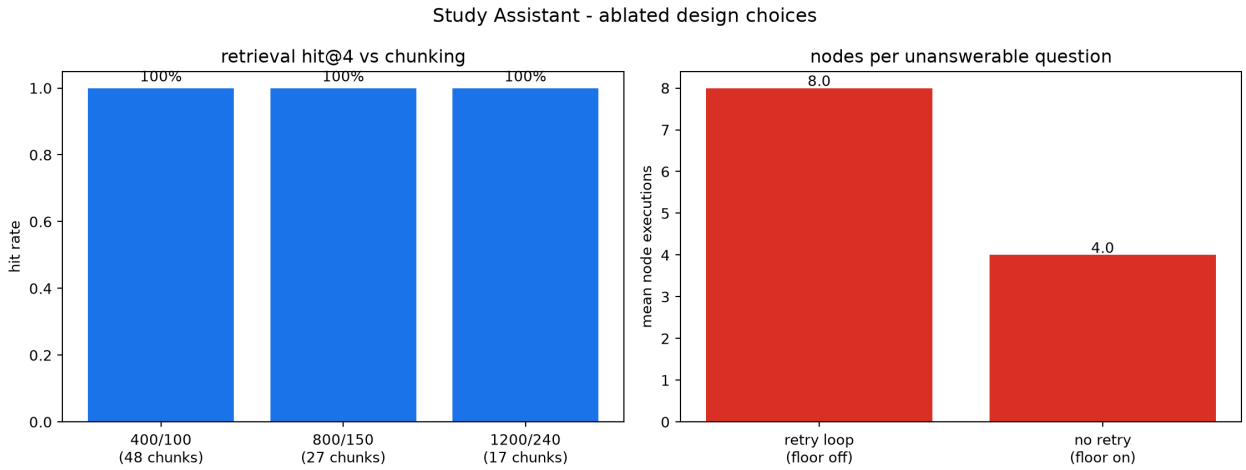


Figure 2. Chunking and retry-policy ablations.

5. Traces

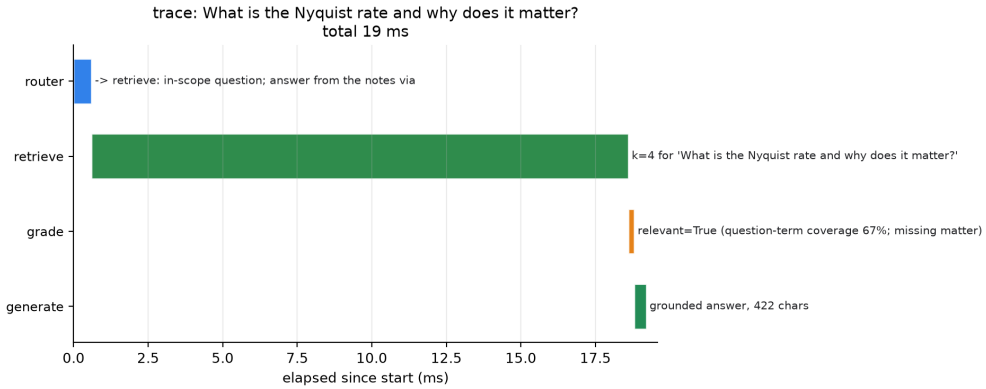


Figure. grounded answer, happy path.

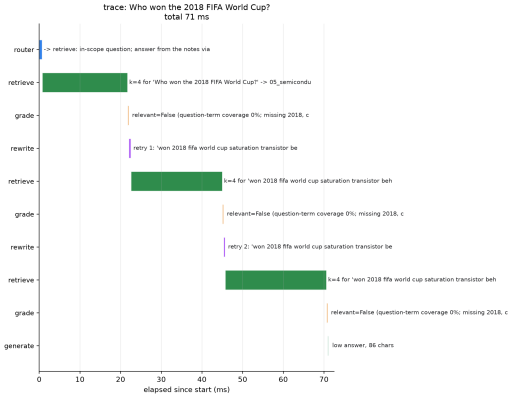


Figure. retry loop (floor off) on an unanswerable question.

6. Demo (4.4)

src/app.py is a Streamlit chat UI showing the answer, a confidence badge, the citations, and an expandable full agent trace and retrieved chunks for every turn, with the live backend and index stats in the sidebar.

7. Limitations

- Small self-authored corpus and test set; the 100% metrics are ceilings, not estimates.
- A lexical relevance grader - the direct cause of the headline failure (q20).
- Extractive default answering; a generative baseline was not run (no key available).
- Single-run measurements; no variance study.

8. Reproduce

python data/build_corpus.py; python src/ingest.py; python eval/run_eval.py; python eval/improvement_experiment.py; streamlit run src/app.py. Set GROQ_API_KEY, GOOGLE_API_KEY or LANGSMITH_API_KEY in .env to enable the hosted model and remote tracing.